BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Strings: Concatenation and Slicing

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. What does assignment do to a variable?

2. What is NOT true?
   A) true
   B) 0.5
   C) false
   D) An empty string

3. What values does a Boolean variable represent?

4. Computers share files on one school site. Two sites in different towns also need a link. Correct both claims. Explain why. (a) A LAN spans several countries. (b) A WAN only works in one room.

Today's objectives

Manipulate strings using concatenation and substring operations. Use the technique in an independent scenario.

Key vocabulary - Read, cover, recall

Concatenation

The + operator joins two strings in order without automatically inserting a space.

Substring

substring(start,length) returns characters beginning at a zero-based index.

Length

The .length property gives the number of characters in a string.

Indexing

The first character in OCR ERL is at position zero.

BLACK PEN: Write one definition in your own words.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New Learning - Key ideas

Concatenation
The + operator joins two strings in order without automatically inserting a space. Apply this in a welcome message.

Substring
substring(start,length) returns characters beginning at a zero-based index. Apply this in a name badge.

Length
The .length property gives the number of characters in a string. Apply this in a club code.

Indexing
The first character in OCR ERL is at position zero. Apply this in a ticket suffix.

I Do - Worked model

A welcome message is being developed. Explain how concatenation and substring are used. [4 marks]

- Read the scenario and identify the data the program needs.
- Concatenation [1]; The + operator joins two strings in order without automatically inserting a space. [1].
- Substring [1]; substring(start,length) returns characters beginning at a zero-based index. [1].
- Two linked points for each concept.

We Do - Guided practice

1. A name badge uses concatenation. Explain how it helps. [2 marks]

2. Explain substring in a club code. [2 marks]

3. A ticket suffix uses length and indexing. Explain both. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Complete the substring to display the first three characters of "NETWORK".

```
name = "NETWORK"
print(name.substring(0, ...))
```

You Do - Independent exam questions

Q2. State TWO relevant techniques from this topic. [2 marks]

Q3. Explain how substring and length help with a school password hint. [4 marks]

Q4. A programmer designs a café menu label. Explain why concatenation, length and indexing matter. [6 marks]

Exit Ticket - True or False?

The + operator joins two strings in order without automatically inserting a space.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. It stores a value in the named variable, replacing its previous value if one existed.

2. C - false

3. True or false.

4. (a) A LAN covers a small area, such as a school site. (b) A WAN links networks over a large area, often using outside providers' links. Internet access alone does not decide the type.

I Do - Explain the model

Read the scenario and identify the data the program needs.

Concatenation [1]; The + operator joins two strings in order without automatically inserting a space. [1].

Substring [1]; substring(start,length) returns characters beginning at a zero-based index. [1].

Two linked points for each concept.

We Do - Model answers

1. Concatenation [1]; The + operator joins two strings in order without automatically inserting a space. [1].

2. Substring [1]; substring(start,length) returns characters beginning at a zero-based index. [1].

3. Length [1]; The .length property gives the number of characters in a string. [1]. Indexing [1]; The first character in OCR ERL is at position zero. [1].

Code First - Answer

```
name = "NETWORK"
print(name.substring(0, 3))
```

Expected output: NET

You Do - Self-mark

Q2: Concatenation [1]; Length [1].

Q3: Substring [1]; substring(start,length) returns characters beginning at a zero-based index. [1]. Length [1]; The .length property gives the number of characters in a string. [1].

Q4: Concatenation [1]; The + operator joins two strings in order without automatically inserting a space. [1]. Length [1]; The .length property gives the number of characters in a string. [1]. Indexing [1]; The first character in OCR ERL is at position zero. [1].

GREEN PEN: Choose the hardest answer. Correct it on lined paper.